Chhatrapati Shahu Ji Maharaj University

Project Synopsis

on

Linux Sever Automation Toolkit

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Bachelors of Computer Application

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ABSTRACT

In modern IT environments, configuring a Linux server is a repetitive task and time-consuming. System administrators must install software packages, configure services, manage users and prepare it for deployment. Manual configuration can lead to errors, inconsistencies, and increased setup time.

The Linux Server Automation Toolkit is a bash-bashed automation solution that simplifies the process of setting up a fresh ubuntu server. The tool performs essential server configuration tasks automatically, ensuring consistency, reducing manual effort, and improving deployment efficiency. It maintains logs of all performed operations.

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1. INTRODUCTION

1.1 Objective

The primary objective of this project is to develop a simple tool which helps developers to configure servers automatically for deployment. This tool aims to:

- Install and configure essential software packages.
- Configure docker and required system services.
- Automate firewall and user management.
- Reduce server setup time and human errors.

1.2 Scope

This project is intended for Ubuntu Server systems and focuses on common server administration tasks. The scope includes:

- Installs and configures essential software such as Docker, Git, and other system utilities.
- Performs system updates and package upgrades automatically.
- Configures basic server security, including user management and UFW firewall settings.
- Uses a configuration file to customize the setup process.

2. SYSTEM REQUIREMENTS

2.1 Software Requirements

- Operating System: Any (if you don't want to use software directly on system).
- Virtualization Software: VirtualBox or VMware Workstation.
- Scripting Language: Bash.
- Version Control: Git.
- Firewall: UFW.
- Text Editor: Visual Studio Code or Nano.

2.2 Hardware Resource Requirements

- Processor: Intel Core i3 or equivalent.
- RAM: Minimum 4GB.
- Storage: Minimum 30 GB free disk space.
- Virtualization Support: Intel VT-x or AMD-V enabled.
- Network: Internet connection for downloading packages and updates.

3. MODULES

3.1 Modules Name

1. System Update and Package Management Module.
2. Docker Installation and Configuration Module.
3. User and Permission Management Module.
4. Security and Firewall Configuration Module.
5. Configuration Management Module.
6. Logging and Error Handling Module.

3.2 Modules Description

- **System Update and Package Management Module:** This module automates system updates and installs required software packages. It checks for available updates, upgrades existing packages, and installs essential utilities required for server setup.
- **Docker Installation and Configuration Module:** This module installs Docker and Docker Compose, starts and enables Docker service, and configures Docker access for users. It prepares the server to run containerized applications.
- **User and Permission Management Module:** This module automates user creation, group assignment, and permission configuration. It ensures proper access control and improves server management.
- **Security and Firewall Configuration Module:** This module configures basic server security by setting up UFW firewall rules, allowing required ports, and applying recommended security settings.
- **Configuration Management Module:** This module reads user defined settings from a configuration file and customizes the provisioning process according to selected options.
- **Logging and Error Handling Module:** This module records all installation and configuration activities with timestamps. It also detects errors and provides meaningful messages during execution.

4. APPLICATION ARCHITECTURE

1. **User:** Updates the configuration file (config.conf) with the required server settings and executes the provisioning script.
2. **Configuration File:** Stores server configuration parameters such as software to install, user details, firewall settings, and other provisioning options.
3. **Provisioning Engine:** Reads the configuration file, validates the settings, and coordinates the execution of all provisioning modules.
4. **Provisioning Modules:** Performs individual tasks such as system update, package installation, Docker installation and configuration, user management, firewall configuration, and logging.
5. **Linux Operating System:** Executes system-level commands using Linux utilities (apt, systemctl, ufw, usermod, etc.) to configure the server.
6. **Logging Module:** Records all provisioning activities, installation status, and errors in log files for monitoring and troubleshooting.
7. **Provisioned Server:** Produces a fully configured Ubuntu server with the required software, services, security settings, and logs.

5. TECHNOLOGY USED

- **Bash Scripting:** Develops the automation scripts for provisioning and configuration.
- **Docker:** Installs and configures the containerization platform on the server.
- **Git:** Version control for managing the project source code.
- **APT Package Manager:** Installs and updates software packages from Ubuntu repositories.
- **Systemd (systemctl):** Manages and controls Linux services.
- U**FW(Uncomplicated Firewall):** Configures basic firewall and network security settings.
- **VirtualBox/ VMware:** Provides a virtual environment for testing the provisioning tool safely.
- **Visual Studio Code/ Nano:** Used to write and edit Bash scripts.
- **Linux Shell Utilities:** Commands such as curl, wget, tar, rsync, chmod, chown, and useradd are used for automation and system configuration.

6. REFERENCES

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2. **Docker Documentation:** Docker Official Documentation. Available at: https://docs.docker.com/
3. **GNU Project:** Bash Reference Manual. Available at: https://www.gnu.org/software/bash/manual/
4. **Git Documentation:** Git Official Documentation. Available at: https://git-scm.com/doc
5. **Ubuntu Manual Pages:** APT Package Manager Documentation. Available at: https://manpages.ubuntu.com/
6. **Ubuntu Documentation:** UFW(Uncomplicated Firewall) Guide. Available at: https://help.ubuntu.com/community/UFW
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